

Emerging Resistance to Novel β -Lactam β -Lactamase Inhibitor Combinations in *Klebsiella pneumoniae* bearing KPC Variants

Mase Hamza^{#1}, German M. Traglia^{#2}, Lucia Maccari³, Sonia Gomez³, Maria Belen Sanz³, Usman Akhtar¹, Vyanka Mezcord¹, Jenny Escalante¹, Alejandra Corso³, Cecilia Rodriguez^{1,4}, Christopher R. Bethel⁵, Gauri G. Rao⁶, Marcelo E. Tolmasky¹, David Paterson⁷, Robert A. Bonomo^{5,8,9*}, Fernando Pasteran³, Maria Soledad Ramirez^{1*}.

¹Center for Applied Biotechnology Studies, Department of Biological Science, College of Natural Sciences and Mathematics, California State University Fullerton, Fullerton, California, USA.

²Unidad de Genómica y Bioinformática, Departamento de Ciencias Biológicas, CENUR Litoral Norte, Universidad de la República, Salto 50000, Uruguay.

³National Regional Reference Laboratory for Antimicrobial Resistance (NRL), Servicio Antimicrobianos, Instituto Nacional de Enfermedades Infecciosas, ANLIS Dr. Carlos G. Malbrán, Buenos Aires, Argentina.

⁴Centro de Referencia para Lactobacilos (CERELA), CONICET, Tucuman, Argentina.

⁵Research Service and GRECC, Louis Stokes Cleveland Department of Veterans Affairs Medical Center, Cleveland, Ohio, USA.

⁶University of Southern California, Los Angeles, USA.

⁷Saw Swee Hock School of Public Health, National University of Singapore, Singapore

⁸Departments of Medicine, Pharmacology, Molecular Biology and Microbiology, Biochemistry, Proteomics and Bioinformatics, Case Western Reserve University School of Medicine, Cleveland, Ohio, USA.

⁹CWRU-Cleveland VAMC Center for Antimicrobial Resistance and Epidemiology (Case VA CARES), Cleveland, Ohio, USA.

[#] Both authors have equal contribution to the work.

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27 ***Corresponding authors:**

28 María Soledad Ramírez, PhD.

29 Professor of the Dept. Biological Science,

30 California State University Fullerton

31 800 N State College Blvd

32

33 Robert A. Bonomo, MD

34 Clinical Scientist Investigator, Veterans Health Administration

35 Medical and Research Services, Louis Stokes Cleveland VA Medical Center

36 Case VA CARES and Departments of Medicine, Molecular Biology and Microbiology, Case

37 Western Reserve University School of Medicine

38 10701 East Blvd., Cleveland, Ohio 44105

39

ABSTRACT

Background: *Klebsiella pneumoniae* carbapenemase (KPC) variants, predominantly KPC-2 and KPC-3, are significant global resistance mechanisms. KPC-2 and KPC-3 confer resistance to a broad range of β -lactams, including carbapenems, while remaining susceptible to ceftazidime-avibactam (CZA). Recently, new KPC variants have developed resistance to CZA through mutations, insertions, or deletions in regions such as the Ω -loop, 240-loop (237–243 aa), and 270-loop (266–275 aa). This study aimed to investigate the collateral resistance to cefiderocol (FDC) and cefepime/zidebactam (FPZ) among isolates with these mutations.

Methods: Fifteen clinical isolates of KPC-producing *Klebsiella spp.* were analyzed, representing 15 distinct variants. Antimicrobial susceptibility testing determined the MICs for CZA, carbapenems, FDC, FPZ, and other antibiotics. Synergy between CZA and FDC was assessed. Whole-genome sequencing (WGS) was used to identify mutations contributing to resistance.

Results: CZA resistance was confirmed in 12 of the 15 KPC variants. Collateral resistance to FDC was observed in eight isolates, with five exhibiting spontaneous resistant subpopulations. Six FDC-resistant strains had mutations in the 270-loop (266–275 aa). Collateral resistance to FPZ was seen in three KPC variants, especially those with mutations in the 270-loop (266–275 aa), though many Ω -loop and 240-loop (237–243 aa) mutants remained susceptible. WGS of FDC-resistant subpopulations revealed additional mutations in *ompC*, *rpoC*, *dksA*, and *cirA*.

Conclusions: This study demonstrates that emerging KPC variants showing resistance to CZA also exhibit resistance to FDC, with collateral resistance to FPZ observed to a lesser extent. Identifying mutations in *bla*_{KPC}, *cirA*, and other genes is important to understand resistance mechanisms for effective therapies.

INTRODUCTION

Klebsiella pneumoniae carbapenemase (KPC) variants, specifically KPC-2 and KPC-3, have gained significant global prevalence, particularly in *K. pneumoniae* isolates [1-3]. These β -lactamases provide resistance to most β -lactams, including carbapenems, but remain susceptible to newer β -lactam/ β -lactamase inhibitors like ceftazidime-avibactam (CZA). However, since the introduction of CZA, numerous KPC variants resistant to this combination have emerged [2, 3]. In these KPC variants, mutations, insertions, and/or deletions have been identified in distinct regions of KPC β -lactamase. Mutational “hot spots” associated with resistance to CZA are located in specific regions: (i) the Ω -loop (residues 164 to 179, which border the lower part of the catalytic pocket), (ii) the 240-loop (amino acids 237 to 243, adjacent to the conserved KTG motif and defining the right side of the active site), and (iii) the 270-loop (amino acids 266 to 275, positioned further from the active site between β strand 5 and the α 11 helix) [3-6]. The emergence of ceftazidime-avibactam resistance in KPC-producing *K. pneumoniae* has been associated with mutations in these critical structural regions. Recent studies have described novel KPC variants, such as KPC-189 and KPC-197, which confer resistance to CZA through modifications in these sites [7, 8]. Additionally, the KPC-74 variant has been identified as a CZA-resistant enzyme that emerged during treatment, highlighting the ongoing evolution of KPC enzymes under selective pressure[9]. Understanding the effects of these substitutions is essential for tracking the evolution and spread of KPC variants resistant to CZA and its collateral resistance to other antibiotics, such as cefiderocol (FDC) and cefepime/zidebactam (FPZ).

Published literature have reported the occurrence of collateral resistance to FDC in few KPC variants, including KPC-31, KPC-33, KPC-62, and novel variants like KPC-109 and KPC-203 (Table S1). The concept of “collateral resistance” has been described in the literature as the unintended resistance to one antibiotic due to selective pressure exerted by another, even when they do not share the same direct target [10, 11]. Specifically, mutations in *bla*_{KPC} associated with CZA resistance have been linked to structural changes in the enzyme that impact its interaction with other β -lactams, including FDC [12]. Similarly, porin alterations—often secondary to β -lactamase evolution—can restrict FDC uptake, further supporting the collateral resistance phenomenon [13]. Most of these variants exhibiting FDC collateral resistance have substitutions in the Ω -loop region, significantly affecting the hydrolysis of both antibiotics and contributing to FDC collateral resistance. Notably, KPC-109, a variant of KPC-3, was identified in a clinical isolate (NE368) with a six-amino acid insertion in the 270-loop region, mediating resistance to CZA and FDC[14].

To our knowledge, collateral resistance to FPZ and cefepime/taniborbactam in clinical isolates has not been reported in the literature. The aim of our present study is to describe the occurrence of collateral resistance to FDC and FPZ in *Klebsiella* clinical isolates harboring various KPC variants (n=15) in the three main specific regions.

MATERIALS and METHODS

Bacterial strains

A total of 37 out of 175 *Klebsiella* spp. clinical isolates harbored a KPC variant distinct from KPC-2 or KPC-3. A representative isolate from each variant was selected for this study. A total of fifteen selected KPC-producing *Klebsiella* spp. clinical isolates, not epidemiologically related and

collected from various regions in Argentina over the span of four years (2020-2023), were included (Table S2). Among these isolates, 15 KPC variants were included (Table 1 and Figure 1). The *K. pneumoniae* ATCC reference strains BAA1705 (KPC-2 producer) and 13883 (susceptible) were also tested as reference.

Antimicrobial susceptibility testing (AST)

The minimum inhibitory concentrations (MICs) for CZA, meropenem (MEM), imipenem (IMP), cefepime (FEP), zidebactam (ZID), FPZ, ertapenem (ETP), ertapenem/zidebactam (ETP/ZID*), imipenem/relebactam (I/R), and meropenem/varbobactam (M/V) were determined using broth microdilution methods and/or commercial E-strips (Liofilchem S.r.l., Roseto degli Abruzzi, Italy) in accordance with the Clinical and Laboratory Standards Institute (CLSI) guidelines. For testing FDC susceptibility, commercial E-strips (Liofilchem S.r.l., Roseto degli Abruzzi, Italy) and broth microdilution with iron-depleted cation-adjusted Mueller-Hinton medium as the reference method were used. All procedures were carried out in accordance with the manufacturer's instructions and met the standards of CLSI [15] and the European Committee on Antimicrobial Susceptibility Testing (EUCAST) (https://www.eucast.org/clinical_breakpoints).

Resistant subpopulations within the inhibition ellipse zones of FDC were selected for further studies and whole genome sequence analysis. Stocks of the FDC resistant subpopulations were stored at -80°C and the stability of the FDC resistance was determined.

For FPZ categorization, the CLSI FEP breakpoint was used. Synergy between CZA and FDC was performed in resistant and heteroresistant strains using MTS™ Synergy Application System (Liofilchem S.r.l., Roseto degli Abruzzi, Italy). Synergy was evaluated using the fractional inhibition concentration index (FICI) as previously described [16]. Quality control strains, such

as *Escherichia coli* ATCC 25922 and the *K. pneumoniae* ATCC 13883 strains were included in the experiments. Each strain was tested at least in duplicates.

Whole genome sequencing analysis

The genomic DNA extraction of parental strain and intra halo colonies was performed using Wizard Promega kit (Promega, Madison, WI, USA). The Genomic sequencing was done using NovaSeq X Plus, producing 2x151 bp paired-end reads. To ensure sequence quality, FASTQC software analysis (<https://www.bioinformatics.babraham.ac.uk/projects/fastqc/>) was performed, followed by trimming and filtering using Trimmomatic software (version: 0.40, ILLUMINACLIP: TrueSeq3-PE.fa.2:30:10; LEADING:3; TRAILING:3; SLIDINGWINDOW: 4:15; MINLEN:36) [17]. De novo sequence assembly was conducted using SPAdes (version: 3.15.4, default parameters) [18] and the quality was subsequently evaluated using QUAST (version: 5.2.0) [19]. Genome annotation was performed through PROKKA (version 1.14.5)[20], while variant calling was carried out using the *breseq* and *gdtools* software packages (version: 0.38.1, consensus mode, default parameters) [21]. Recombination regions were identified and removed using Gubbins software (version: 3.3.0, default parameters) [22]. Plasmid identification was performed using PlasmidFinder v2.1 [23]. The copy number gene (*bla_{KPC}*) was assessed using CCNE tool [24]. The Whole Genome Shotgun project of KPNMA215, KPNMA216 and KAMA222 strains has been deposited in GenBank under accession numbers JBIUGI0000000000, JBIUGH0000000000, and JBIUGG0000000000 respectively. In addition, the fastq files, quality analysis files, assemblies' sequences and genome annotation files of wild-type and IHC were uploaded in Zenodo repository: <https://zenodo.org/records/15122353>(last accessed April 1, 2025).

RESULTS AND DISCUSSION

Description of CZA resistance and collateral resistance to FDC in *Klebsiella* KPC variants strains

The CZA MICs for 15 KPC variants were determined, confirming resistance in 12 of them (Table 1). Interestingly, three variants with mutations in the Ω -loop—KPC-25, KPC-160, and KPC-165—remained susceptible to CZA, despite alterations in this critical region (Table 1). Prior publications of these specific KPC variants have been identified in the literature.

We next studied the collateral resistance to FDC, and we observed resistance in eight of the 15 isolates, using EUCAST breakpoint (BP) guidelines. Among the tested strains, resistant colonies with the gradient strip inhibition zone (FDC resistant subpopulations) were also observed in five strains including one that was categorized as FDC susceptible (Table 1). Six of the FDC resistant strains harbor substitutions in the 266–275 loop (Table 1). Previous reports have identified cross resistance to FDC when substitutions in the Ω -loop are present (Table S1) [3]. In addition, these studies highlight the emergence of FDC resistance in high-risk clones such as the ST307 lineage, which has been responsible for outbreaks in hospital settings, particularly in ICUs [25, 26]. KPC-62, identified in ST307 isolates, displayed resistance to both CZA and FDC, driven by the L169Q mutation within the *bla*_{KPC} gene. The novel KPC-203 variant found in Italy adds further complexity to the resistance landscape, as it exhibits cross resistance to CZA and FDC due to significant modifications at key positions, including a deletion in the Ω -loop and an insertion in the 260 amino acid position [27]. The newly identified KPC-216 shows cross-resistance to both CZA and FDC. The KPC-216 variant, characterized by a lysine insertion at position 170 in the Ω -loop, was isolated from a *K. pneumoniae* ST101 strain and demonstrated resistance to both CZA and FDC [28]. In addition, KPC-109, a variant of KPC-3, was identified in

the clinical isolate, featuring a six-amino acid insertion in the 270-loop region, which conferred resistance to both CZA and FDC [14]. Reports with mutations such as the D179Y substitution found in both KPC-31 and KPC-33, and observed in this work, illustrate the challenge of treating infections with these variants, as they lead to not only CZA resistance but also a potential for cross-resistance with FDC [29-32] .

We tested the synergy between CZA and FDC in all strains exhibiting resistance or heteroresistance to both antibiotics. Synergy was observed in four strains (Table 2). Particularly, synergy was seen in the strains exhibiting intracolony for FDC (Table 1 and Table 2). Evaluating the synergy between FDC and CZA, a clinically available fixed combination, addresses the need to explore relevant therapeutic options. While avibactam has previously demonstrated synergy with FDC, it is not available as a stand-alone agent for clinical use. In contrast, CZA is a formulation used in clinical practice that contains both ceftazidime and avibactam. Therefore, assessing the synergy between CZA and FDC allows for a more translationally relevant interpretation of potential combination therapies.

Impact of KPC variants in cefepime/zidebactam susceptibility

Firstly, we observed that all 15 KPC variants were resistant to cefepime (FEP), and all but one were resistant to zidebactam (ZID) (Table 1). When testing FPZ, three of the KPC variants displayed resistance (Table 1). Notably, most strains harboring mutations in the Ω -loop and the 240-loop remained susceptible to FPZ.

Only one report on FPZ has been published, where the authors evaluated the antimicrobial susceptibility of a KPC-33 variant, which exhibited susceptibility to FPZ (Shi Qingyu, 2020, DOI: 10.1093/CID/CIAA1521). Another report evaluated

cefepime/taniborbactam, showing decreased susceptibility in laboratory-constructed strains with substitutions in both the Ω -loop and the 240-loop, particularly when combined with porin defects (mutations in OmpK35 and OmpK36) [33]. Our study is the first to describe collateral resistance to FPZ in KPC variants.

When compared with KPC-producing clinical isolates from the same period that lacked KPC mutations, variants exhibiting resistance to CZA were significantly more likely to be resistant to FPZ and FDC and more susceptible to carbapenem (Fisher's exact test, $p < 0.05$) [34]. While based on limited samples, this comparison and analysis fit for normal distribution and open new venues for future studies with larger cohorts.

As amino acid changed in the MarR, PBP-2, PBP-3, OmpK35, and OmpK36 proteins, or their respective promoters, may contribute to the FDC and FPZ resistance phenotypes [35, 36]. We analyzed the sequences of the three KPC-producing strains with FPZ resistance. A comparative analysis of KPNAMA215 and KPNAMA216 against the FDC-susceptible ATCC 13883 strain revealed no amino acid substitutions in PBP-2 and PBP-3. However, in OmpK35, only KPNAMA216 exhibited an A8T substitution, while multiple amino acid changes were observed in OmpK36 in both KPNAMA215 and KPNAMA216 (Fig. S1).

The comparison of the KAMA222 genome to *K. aerogenes* FDAARGOS 1442 (used as the reference genome) identified two amino acid substitutions in PBP-2 (G107D and A354D) (Fig. S2), as well as multiple changes in OmpK36, consistent with the findings in KPNAMA215 and KPNAMA216 (Fig. S1 and Fig. S2). In MarR, an S82G amino acid substitution was detected in KPNAMA215 and KPNAMA216, whereas KAMA222 showed no amino acid changes in this protein (Fig. S2). The presence of multiple amino acid substitutions in OmpK36 across the

analyzed genomes complicates the assessment of their specific contribution to FDC and FPZ resistance. However, we cannot rule out the possibility that these alterations play a role in the observed resistance. Further analysis involving a larger number of strains and their respective isogenic strain with not mutation on the porins is necessary to draw definitive conclusions. In conclusion, while previous reports have highlighted susceptibility to FPZ in specific KPC variants, such as KPC-33, and documented reduced susceptibility in other FEP combinations like FEP/taniborbactam under certain genetic conditions, our study is the first to describe collateral resistance to FPZ in multiple KPC variants. This finding underscores the complexity of resistance mechanisms in KPC-producing strains, particularly those involving mutations in the 270-loop (266-275 aa).

Collateral resistance to ertapenem/zidebactam (ETP/ZID) KPC variants

Among the 15 KPC variant isolates tested in this study, six were resistant to ertapenem (ETP) (Table 1). When evaluating the ETP/ZID combination, susceptibility was restored in only two isolates (Table 1). Most of the ETP-resistant strains harbored mutations in the 270-loop (266-275 aa), which may contribute to their resistance phenotype (Table 1).

Notably, one isolate (KPNMA216) exhibited the highest MIC (>32 mg/L) to ETP and ETP/ZID, and this strain possessed alterations in both OmpK35 and OmpK36 and the 270-loop (266-275 aa) mutations on the KPC (Table 1 and Fig. S1). The presence of porins mutations suggests a potential role in limiting β -lactam uptake and enhancing resistance to β -lactam/ β -lactamase inhibitor combinations.

Although all the isolates analyzed in this study come from Argentina, the resistance mechanisms identified, in particular the mutations in *bla*_{KPC} associated with resistance to CZA

and its impact on FDC and FPZ, have been reported in various regions of the world, including in high-risk clones such as ST258 and ST307 [7, 25, 37, 38]. Our findings agree with these previous studies, suggesting that the described mechanisms can be extrapolated beyond the local context.

To date, there have been no prior reports of collateral resistance to FDC and ETP/ZID in KPC-producing isolates [39-41]. This study showed the occurrence of this phenomenon for the first time and underscores the critical need for ongoing comprehensive surveillance to monitor emerging resistance patterns in multidrug-resistant *Klebsiella*.

Collateral susceptibility to carbapenems in *K. pneumoniae* KPC variants

We tested the MICs of imipenem (IMP) and meropenem (MEM) in the 15 KPC variants, and we observed in 12 strains collateral susceptibility to IMP (n=1), MEM (n=5), and both (n=6) (Table 1). MEM exhibits lower MICs compared to imipenem, indicating that it is generally less affected by resistance mechanisms, including *bla*_{KPC} variants. Cross resistance to carbapenems was seen in strains harboring KPC variants with substitutions in the 270-loop (266-275 aa) (Table 1). The mutations in the 270-loop region can alter the local flexibility and conformation of the active site, influencing substrate binding and catalytic efficiency, particularly in class A β -lactamases [42-44].

Previous investigations have suggested that certain substitutions in KPC variants, particularly those in the Ω -loop region, may reduce the enzyme's carbapenemase activity, thereby restoring susceptibility to carbapenems [3, 4, 45, 46]. For instance, variants such as KPC-31 and KPC-33, which harbor the D179Y substitution, have been shown to confer resistance to CZA while exhibiting collateral susceptibility to carbapenems. No isolates with mutations in 270-loop

(266-275 aa) showed collateral susceptibility to imipenem. Although our study did not include kinetic validation, previous literature has demonstrated the functional relevance of these mutations through mechanistic approaches, showing that substitutions, insertions or deletions can contribute to CZA resistance by increasing affinity for CAZ and reducing susceptibility to AVI, ultimately affecting enzyme stability and broadening its spectrum of activity [47] [3, 9]. In addition, we tested imipenem/relebactam (I/R) and meropenem/varbobaactam (M/V) and we observed that vaborbactam reduced meropenem MICs by more than 2 log₂-fold in 12 strains, whereas relebactam achieved the same reduction in 8 out of 15 strains. (Table 1).

Our work and previous findings highlight the complex interplay between β -lactamase mutations and antibiotic efficacy, where changes that drive resistance to one class of antibiotics may inadvertently restore susceptibility to others, such as carbapenems. More research is needed to further explore this phenomenon across different KPC variants.

WGS analysis of FDC spontaneous emergent resistant cells

Spontaneous hetero-resistant colonies from three KPC variants strains were chosen for further analysis. The MIC for FDC for these resistant subpopulations showed a 2-fold increase for KPNMA215 and KPNMA216 and a 3-fold increase KPNMA222 (Table 3).

The WGS was performed and subsequent genome analyses of *K. pneumoniae* strains KPNMA215 and KPNMA216, along with *K. aerogenes* strain KAMA222 and their FDC-resistant subpopulations, revealed distinct sequence types (STs) and species classifications. KPNMA215 was classified as ST15, KPNMA216 as ST14, and KAMA222 as ST92. Common antimicrobial resistance (AMR) genes were found across all strains, including *bla*_{TEM}, *bla*_{SHV}, *oqx*A, *oqx*B, *fos*A6, *tet*(A), and *tet*(D), as well as aminoglycoside-

modifying enzymes such as *aac(6')-Ib-cr* and *aph(6)-Id* (Table S3). The presence of plasmids was identified in all three sequenced genomes. All three genomes harbor plasmids belonging to the incompatibility group IncL/M. Additionally, an IncFII-type plasmid was found in KPNAMA215 and KPNAMA222. An IncFIB-type plasmid was identified in the genomes of KPNAMA215 and KPNAMA216. Furthermore, KPNAMA215 also carries a plasmid belonging to the incompatibility group IncR. No resistance genes were identified on the plasmids.

In the mutational analysis of the FDC-resistant subpopulation IHC215, a mutation was detected in the *ISSod9* transposase, which plays a role in the movement of genetic elements and may facilitate the horizontal transfer of resistance genes. Additionally, a mutation was found in the *rpoC* gene, which encodes the β -subunit of DNA-directed RNA polymerase. Mutations in *rpoC* have been associated with transcriptional accuracy and stress adaptation, which may contribute to bacterial survival under antibiotic pressure by modulating global gene expression. The mutation identified in *rpoC* in IHC215 may alter RNA polymerase function and global gene expression patterns, ultimately reducing susceptibility to FDC (Table 3 and Table S4).

For KPNAMA216 FDC-resistant subpopulation IHC216, mutations were detected in the *ompC* gene. This porin facilitates passive diffusion of β -lactams into the periplasmic space, and its alteration likely reduces permeability, limiting antibiotic entry and contributing to resistance. Similar resistance mechanisms have been observed in clinical isolates involving OmpK35 and OmpK36 porins in *K. pneumoniae* [27, 33, 48]. The presence of the *ompC* mutation may contribute to the reduced susceptibility to antibiotics [33]. A mutation in *dksA* gene, which encodes an RNA polymerase-binding protein, was also observed, suggesting its role

in regulating gene expression under stress, particularly during antibiotic exposure (Table 3 and Table S4) [49]. Specifically, a deletion of the first 93 nucleotides was identified in the *dkcA* gene, potentially leading to a loss-of-function mutation. This truncation may impair the bacterial stress response and further modulate the expression of resistance or persistence-related genes (Table S4). Lastly, in the IHC222 subpopulation, we identified a typical mutation in the *cirA* gene, a gene frequently prone to mutations and previously linked to FDC resistance (Table 3 and Table S4). *cirA* encodes a TonB-dependent outer membrane receptor responsible for ferric-siderophore uptake. Although the mutation identified here has not been previously reported, its location and nature suggest functional inactivation. The novelty of this mutation highlights the genetic flexibility of *cirA* and its role as a resistance determinant. A previous case study of a hypervirulent *K. pneumoniae* strain co-producing KPC-2 and SHV-12 identified a truncation in *cirA*, leading to FDC resistance [50]. Similarly, other studies have shown that *cirA* mutations, when combined with the production of NDM-5 carbapenemase, result in even higher levels of FDC resistance [26, 51-54]. This synergy between *cirA* inactivation and β -lactamase activity underscores the complex interplay of resistance mechanisms. Previous findings and our current work emphasize the need to monitor *cirA* mutations as a significant marker of FDC resistance in clinical settings, particularly in strains producing KPC-2, and KPC variants, as well as NDM-5.

The copy number of the *bla*_{KPC} gene and the associated increase in carbapenems, CZA, and β -lactamase/inhibitors resistance have been previously reported [35, 48, 55]. In this study, the copy number of the *bla*_{KPC} gene in both WT and IHC colonies of strains KPNAMA215, KPNAMA216, and KAMA222 was evaluated using whole genome sequencing data and the CCNE

tool. An increased *bla*_{KPC} copy number was observed in the IHC colonies of KPNAMA215 (1.52-fold changes) and KAMA222 (1.48-fold changes) compared to their respective WT colonies. In contrast, no difference in the *bla*_{KPC} copy number was detected in the IHC colonies of KPNAMA216.

Together, our findings indicate that mutations in *cirA*, *ompC*, *rpoC*, and *dksA* play complementary roles in reducing susceptibility to FDC by affecting membrane permeability, drug uptake, and transcriptional regulation under stress conditions. These mechanisms highlight the multifactorial nature of resistance and the importance of monitoring such mutations in clinical surveillance.

A limitation of our study is that, although we identified mutations potentially associated with resistance, such as in the *ompC* and *rpoC* genes, we did not experimentally validate their functional impact using gene knockout or complementation assays. Although these experiments would provide functional confirmation, our primary objective was to identify mutational events potentially related to FDC and CZA resistance in clinical isolates with KPC variants, employing genomic surveillance and comparative sequence analysis. Nevertheless, our findings underscore the need for future research to determine the functional impact of these mutations on the resistance phenotype.

Conclusion: This study highlights the complex genetic landscape of KPC variants and the emergence of collateral resistance to both FDC, FPZ and ETP/ZID. Notably, several variants, particularly those harboring mutations in critical regions such as the Ω -loop and the 270-loop, contributed to high levels of resistance to CZA and FDC. Collateral resistance to FPZ was observed to a lesser extent, specifically among variants with mutations in the 270-loop. The

detection of collateral resistance to both antibiotics in strains that had not been previously exposed to these drugs is particularly concerning, as it suggests the potential for resistance to develop even without selective pressure from these antibiotics. Additionally, our findings revealed spontaneous mutations in FDC-resistant subpopulations, including previously known mutations in the *cirA* gene as well as novel mutations not previously associated with FDC resistance. This underscores the importance of continuous monitoring of KPC variants, especially in clinical settings, to prevent further spread of resistance and to develop more effective treatment strategies as KPC allelic variants may influence susceptibility profiles, particularly regarding their response to novel β -lactam agents. Understanding the molecular mechanisms driving resistance will be crucial for managing and controlling the spread of these multidrug-resistant pathogens.

Author Contributions: GMT, GGR, MET, RAB, FP, and MSR conceived the study and designed the experiments. MH, GMT, LM, SG, MBS, CR, UA, VK, JE, FP, and MSR performed the experiments and genomics and bioinformatics analyses. GMT, FP, CRB, GGR, DP, MET, RAB, and MSR analyzed the data and interpreted the results. GMT, FP, MET, and MSR contributed reagents/materials/analysis tools. GMT, FP, GGR, CRB, AC, MET, RAB, and MSR wrote and revised the manuscript. All authors read and approved the final manuscript.

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Conflicts of Interest: The authors declare no conflict of interest.

Legends to Figure and Tables

Figure 1: KPC variants aligned and compared against KPC-2. Protein sequences were retrieved from the CARD database and β -Lactamase Database. Aligned with MUSCLE, visualized by NCBI MSA viewer.

Table 1. KPC variants strains, molecular and phenotypic characteristics.

Strain ID	KPC variant	bla KPC specific region	Mutation detected	Minimum Inhibitory Concentration (mg/L)										Hetero resistance	
				CZA	FDC	FEP	ZID	FPZ	IMP	MEM	ETP	ETP/ZID#	I/R		M/V
KPNMA212	KPC-14	Loop 240	del_242-243_GT	64	0.75	>64	>64	1	0.75	0.125	0.19*	0.125*	0.19	0.064	NO
KPNMA227	KPC-96		Y241N	8	0.5	>512	>64	1	0.5	2	1	No growth	0.094	0.25	YES
KPNMA214	KPC-161	Loop 270	ins 270 - 280_DDKHSEAVIAA	24	6	>512	>64	16	16	8	2*	0.19*	0.75	1	NO
KPNMA215	KPC-162		ins 264 - 270_TRAPNKD	24	4	>512	>64	4	3	1	0.75	0.5	0.094	0.5	YES
KPNMA216	KPC-163		ins 269 - 274_KDDKHS	24	8	>512	>64	16	64	32	>32	>32	2	2	YES
KPNMA217	KPC-164		ins 275 - 277_EAV	24	4	>512	>64	4	8	1	2	0.5	0.032	0.25	YES
KPNMA221	KPC-44		ins 261_AVTAPNKDDKHSE	48	12	512	>64	4	12	4	1.5	1.5	1	0.5	NO
KPNMA225	KPC-80		ins 267 - 270_PNK	48	0.19	>512	>64	2	4	1	0.38	No growth	0.19	0.032	NO
KPNMA228	KPC-97		ins 277 - 281_VNSEA	64	24	32	>64	2	16	2	0.75	0.75	1	0.19	NO
KPNMA213	KPC-160		del 166 - 167_EL	2	0.38	>64	1	1	0.75	0.75	0.75	0.75	0.25	0.064	NO
KPNMA218	KPC-165	Ω-loop	N170D	0.5	<0.016	512	>64	4	0.25	0.064	0.047	No growth	0.125	0.023	NO
KPNMA219	KPC-25		ins 166 - 167_EL	0.5	<0.016	512	>64	0.25	0.5	0.023	0.094*	0.016*	0.125	0.023	NO
KPNMA220	KPC-33		D179Y	32	12	64	>64	2	0.5	4	12	12	0.25	0.75	NO
KAMA222	KPC-57		D179V	64	6	>512	>64	2	3	0.75	0.25	0.25	0.19	0.25	YES
KPNMA224	KPC-73	Combinations of regions		48	0.75	512	>64	16	0.125	2	4*	4*	0.19	0.38	NO
			(del 166 - 167_EL) (ins 269 - 274_KDDKHS) (H273Y)												
										* Intracolony	# 4 ug/ml				

Abbreviations: CZA, ceftazidime avibactam; FDC, cefiderocol; FEP: cefepime; ZID: zidebactam; FPZ: cefepime zidebactam; IMP: imipenem; MEM: meropenem, ETP: ertapenem, I/R: IMP/relebactam, M/V: MEM/varbobactam.

Green cells indicate MIC values that correspond to susceptibility according to CLSI standards. Yellow cells represent values in the Susceptible-Dose Dependent category, while red cells correspond to MIC values classified as resistant according to CLSI criteria

Table 2. Synergy between CZA and FDC for FDC resistant/hetero-resistant KPC variant strains.

MTS synergy				
Strains	MIC CZA(FDC)	MIC FDC(CZA)	FICI	Synergy
KPNMA214	6	2	0.58	Additive
KPNMA215	4	0.5	0.29	Synergy
KPNMA216	6	1	0.38	Synergy
KPNMA217	0.75	0.38	0.13	Synergy
KPNMA220	16	4	0.83	Additive
KPNMA221	64	4	1.67	Additive
KAMA222	4	0.094	0.08	Synergy
KPNMA227	3	0.19	0.76	Additive
KPNMA228	32	8	0.83	Additive

FDC: cefiderocol, FICI: Fractional Inhibitory concentration index.

Synergy was defined as $FICI \leq 0.5$, Additive effect as FICI between 0.5 and 1.

Table 3. Parental and resistant subpopulations FDC MIC and ST, and relevant mutations associated with antibiotic resistance.

FDC MIC (mg/L)				Relevant mutations				ST
IHC Strains	KPC variants	Parental	Intracolony	<i>ompC</i>	<i>dkcA</i>	<i>cirA</i>	<i>rpoC</i>	
KPNMA215	KPC-162	4	16	wild-type	wild-type	wild-type	R220C (CGC→IGC)	ST15
KPNMA216	KPC-163	8	32	311K (TAG→AAG)	deletion (55/456 nt)	wild-type	wild-type	ST14
KAMA222	KPC-57	6	48	wild-type	wild-type	G75R (GGC→CGC)	wild-type	ST92

FDC: cefiderocol, ST: sequence type

Supplementary Material

Figure S1.

Comparison of the OmpK35, OmpK36, and MarR amino acid sequences between KPNMA215, KPNMA216, and *Klebsiella pneumoniae* ATCC 13883, using the latter as the reference genome from the NCBI Genome database.

Figure S2.

Comparison of the OmpK36 and PBP-2 amino acid sequences between KAMA222 and *Klebsiella aerogenes* FDAARGOS 1445, using the latter as the reference genome from the NCBI Genome database.

Table S1: Reported cases in the literature highlight the occurrence of collateral resistance to FDC in KPC variants

Table S2: Demographic data and characteristics of the fifteen *Klebsiella* spp. strains included in this manuscript.

Table S3: Antimicrobial resistance genes present in KPNMA215, KPNMA216 and KAMA222.

Table S4: Mutational analysis of KPNMA215, KPNMA 216, and KAMA222 IHCs compared to the corresponding parental strain.

References

- [1] van Duin D, Doi Y. The global epidemiology of carbapenemase-producing Enterobacteriaceae. *Virulence*. 2017;8:460-9.
- [2] Queenan AM, Bush K. Carbapenemases: the versatile beta-lactamases. *Clin Microbiol Rev*. 2007;20:440-58, table of contents.
- [3] Hobson CA, Pierrat G, Tenaillon O, Bonacorsi S, Bercot B, Jaouen E, et al. *Klebsiella pneumoniae* Carbapenemase Variants Resistant to Ceftazidime-Avibactam: an Evolutionary Overview. *Antimicrob Agents Chemother*. 2022;66:e0044722.
- [4] Taracila MA, Bethel CR, Hujer AM, Papp-Wallace KM, Barnes MD, Rutter JD, et al. Different Conformations Revealed by NMR Underlie Resistance to Ceftazidime/Avibactam and Susceptibility to Meropenem and Imipenem among D179Y Variants of KPC beta-Lactamase. *Antimicrob Agents Chemother*. 2022;66:e0212421.
- [5] Alsenani TA, Viviani SL, Papp-Wallace KM, Bonomo RA, van den Akker F. Exploring avibactam and relebactam inhibition of *Klebsiella pneumoniae* carbapenemase D179N variant: role of the Omega loop-held deacylation water. *Antimicrob Agents Chemother*. 2023;67:e0035023.
- [6] Alsenani TA, Viviani SL, Kumar V, Taracila MA, Bethel CR, Barnes MD, et al. Structural Characterization of the D179N and D179Y Variants of KPC-2 beta-Lactamase: Omega-Loop Destabilization as a Mechanism of Resistance to Ceftazidime-Avibactam. *Antimicrob Agents Chemother*. 2022;66:e0241421.
- [7] Zhang X, Xie Y, Zhang Y, Lei T, Zhou L, Yao J, et al. Evolution of ceftazidime-avibactam resistance driven by mutations in double-copy *bla*(KPC-2) to *bla*(KPC-189) during treatment of ST11 carbapenem-resistant *Klebsiella pneumoniae*. *mSystems*. 2024;9:e0072224.
- [8] De la Cadena E, Mojica MF, Rojas LJ, Castro BE, Garcia-Betancur JC, Marshall SH, et al. First report of KPC variants conferring ceftazidime-avibactam resistance in Colombia: introducing KPC-197. *Microbiol Spectr*. 2024;12:e0410523.
- [9] Li X, Quan J, Ke H, Wu W, Feng Y, Yu Y, et al. Emergence of a KPC Variant Conferring Resistance to Ceftazidime-Avibactam in a Widespread ST11 Carbapenem-Resistant *Klebsiella pneumoniae* Clone in China. *Front Microbiol*. 2021;12:724272.
- [10] Zwep LB, Haakman Y, Duisters KLW, Meulman JJ, Liakopoulos A, van Hasselt JGC. Identification of antibiotic collateral sensitivity and resistance interactions in population surveillance data. *JAC Antimicrob Resist*. 2021;3:dlab175.
- [11] Mahmud HA, Wakeman CA. Navigating collateral sensitivity: insights into the mechanisms and applications of antibiotic resistance trade-offs. *Front Microbiol*. 2024;15:1478789.
- [12] Haidar G, Clancy CJ, Shields RK, Hao B, Cheng S, Nguyen MH. Mutations in *bla*(KPC-3) That Confer Ceftazidime-Avibactam Resistance Encode Novel KPC-3 Variants That Function as Extended-Spectrum beta-Lactamases. *Antimicrob Agents Chemother*. 2017;61.
- [13] Moon SH, Huang E. Cefiderocol Resistance in *Klebsiella pneumoniae* Is Linked to SHV Extended-Spectrum beta-Lactamase Activities and Functional Loss of the Outer Membrane Porin OmpK35. *Microbiol Spectr*. 2023;11:e0349622.
- [14] Di Pilato V, Codda G, Niccolai C, Willison E, Wong JLC, Coppo E, et al. Functional features of KPC-109, a novel 270-loop KPC-3 mutant mediating resistance to avibactam-based beta-lactamase inhibitor combinations and cefiderocol. *Int J Antimicrob Agents*. 2024;63:107030.
- [15] Institute CaLS. CLSI M100-S34 guidelines. CLSI; 2024.

- [16] Gomara M, Ramon-Garcia S. The FICI paradigm: Correcting flaws in antimicrobial in vitro synergy screens at their inception. *Biochem Pharmacol*. 2019;163:299-307.
- [17] Bolger AM, Lohse M, Usadel B. Trimmomatic: a flexible trimmer for Illumina sequence data. *Bioinformatics*. 2014;30:2114-20.
- [18] Bankevich A, Nurk S, Antipov D, Gurevich AA, Dvorkin M, Kulikov AS, et al. SPAdes: a new genome assembly algorithm and its applications to single-cell sequencing. *J Comput Biol*. 2012;19:455-77.
- [19] Gurevich A, Saveliev V, Vyahhi N, Tesler G. QUAST: quality assessment tool for genome assemblies. *Bioinformatics*. 2013;29:1072-5.
- [20] Seemann T. Prokka: rapid prokaryotic genome annotation. *Bioinformatics*. 2014;30:2068-9.
- [21] Deatherage DE, Barrick JE. Identification of mutations in laboratory-evolved microbes from next-generation sequencing data using breseq. *Methods Mol Biol*. 2014;1151:165-88.
- [22] Croucher NJ, Page AJ, Connor TR, Delaney AJ, Keane JA, Bentley SD, et al. Rapid phylogenetic analysis of large samples of recombinant bacterial whole genome sequences using Gubbins. *Nucleic Acids Res*. 2015;43:e15.
- [23] Carattoli A, Zankari E, Garcia-Fernandez A, Voldby Larsen M, Lund O, Villa L, et al. In silico detection and typing of plasmids using PlasmidFinder and plasmid multilocus sequence typing. *Antimicrob Agents Chemother*. 2014;58:3895-903.
- [24] Jiang J, Chen L, Chen X, Li P, Xu X, Fowler VG, Jr., et al. Carbapenemase-Encoding Gene Copy Number Estimator (CCNE): a Tool for Carbapenemase Gene Copy Number Estimation. *Microbiol Spectr*. 2022;10:e0100022.
- [25] Castillo-Polo JA, Hernandez-Garcia M, Morosini MI, Perez-Viso B, Soriano C, De Pablo R, et al. Outbreak by KPC-62-producing ST307 *Klebsiella pneumoniae* isolates resistant to ceftazidime/avibactam and cefiderocol in a university hospital in Madrid, Spain. *J Antimicrob Chemother*. 2023;78:1259-64.
- [26] Hong H, Fan L, Shi W, Zhu Y, Liu P, Wei D, et al. Overexpression of beta-lactamase genes (*bla*(KPC), *bla*(SHV)) and novel *CirA* deficiencies contribute to decreased cefiderocol susceptibility in carbapenem-resistant *Klebsiella pneumoniae* before its approval in China. *Antimicrob Agents Chemother*. 2024;68:e0075424.
- [27] Amadesi S, Bianco G, Secci B, Fasciana T, Boattini M, Costa C, et al. Complete Genome Sequence of a *Klebsiella pneumoniae* Strain Carrying Novel Variant *bla*(KPC-203), Cross-Resistant to Ceftazidime/Avibactam and Cefiderocol, but Susceptible to Carbapenems, Isolated in Italy, 2023. *Pathogens*. 2024;13.
- [28] Giufre M, Errico G, Del Grosso M, Pagnotta M, Palazzotti B, Ballardini M, et al. Detection of KPC-216, a Novel KPC-3 Variant, in a Clinical Isolate of *Klebsiella pneumoniae* ST101 Co-Resistant to Ceftazidime-Avibactam and Cefiderocol. *Antibiotics (Basel)*. 2024;13.
- [29] Zhang P, Wang J, Li Y, Shi W, Cai H, Yang Q, et al. Emergence of *bla*(KPC-33)-harboring Hypervirulent ST463 *Pseudomonas aeruginosa* Causing Fatal Infections in China. *J Infect*. 2022;85:e86-e8.
- [30] Gong G, Chen Q, Luo J, Wang Y, Li X, Zhang F, et al. Characteristics of a ceftadine/avibatam resistance KPC-33-producing *Klebsiella Pneumoniae* strain with capsular serotype K19 belonging to ST15. *J Glob Antimicrob Resist*. 2023;35:159-62.
- [31] Faccone D, de Mendieta JM, Albornoz E, Chavez M, Genero F, Echegorry M, et al. Emergence of KPC-31, a KPC-3 Variant Associated with Ceftazidime-Avibactam Resistance, in an

Extensively Drug-Resistant ST235 *Pseudomonas aeruginosa* Clinical Isolate. Antimicrob Agents Chemother. 2022;66:e0064822.

[32] Birgy A, Nnabuife C, Palzkill T. The mechanism of ceftazidime and cefiderocol hydrolysis by D179Y variants of KPC carbapenemases is similar and involves the formation of a long-lived covalent intermediate. Antimicrob Agents Chemother. 2024;68:e0110823.

[33] Satapoomin N, Dulyayangkul P, Avison MB. *Klebsiella pneumoniae* Mutants Resistant to Ceftazidime-Avibactam Plus Aztreonam, Imipenem-Relebactam, Meropenem-Vaborbactam, and Cefepime-Taniborbactam. Antimicrob Agents Chemother. 2022;66:e0217921.

[34] Echegorry M, Marchetti P, Sanchez C, Olivieri L, Faccone D, Martino F, et al. National Multicenter Study on the Prevalence of Carbapenemase-Producing *Enterobacteriaceae* in the Post-COVID-19 Era in Argentina: The RECAPT-AR Study. Antibiotics (Basel). 2024;13.

[35] Gaibani P, Bianco G, Amadesi S, Boattini M, Ambretti S, Costa C. Increased bla(KPC) Copy Number and OmpK35 and OmpK36 Porins Disruption Mediated Resistance to Imipenem/Relebactam and Meropenem/Vaborbactam in a KPC-Producing *Klebsiella pneumoniae* Clinical Isolate. Antimicrob Agents Chemother. 2022;66:e0019122.

[36] Yousefi B, Kashanipoor S, Mazaheri P, Alibabaei F, Babaeizad A, Asli S, et al. Cefiderocol in Combating Carbapenem-Resistant *Acinetobacter baumannii*: Action and Resistance. Biomedicines. 2024;12.

[37] Arcari G, Cecilia F, Oliva A, Polani R, Raponi G, Sacco F, et al. Genotypic Evolution of *Klebsiella pneumoniae* Sequence Type 512 during Ceftazidime/Avibactam, Meropenem/Vaborbactam, and Cefiderocol Treatment, Italy. Emerg Infect Dis. 2023;29:2266-74.

[38] Blanco-Martin T, Gonzalez-Pinto L, Aja-Macaya P, Rodriguez-Pallares S, Sanchez-Pena L, Gato E, et al. Mutant prevention concentrations, in vitro resistance evolution dynamics, and mechanisms of resistance to imipenem and imipenem/relebactam in carbapenem-susceptible *Klebsiella pneumoniae* isolates showing ceftazidime/avibactam resistance. Antimicrob Agents Chemother. 2024;68:e0112024.

[39] Mushtaq S, Vickers A, Chaudhry A, Woodford N, Livermore DM. Inoculum effects of cefepime/zidebactam (WCK 5222) and ertapenem/zidebactam (WCK 6777) for Enterobacterales in relation to beta-lactamase type and enhancer effect, as tested by BSAC agar dilution. J Antimicrob Chemother. 2022;77:1916-22.

[40] Gethers M, Chen I, Abdelraouf K, Nicolau DP. In vivo efficacy of WCK 6777 (ertapenem/zidebactam) against carbapenemase-producing *Klebsiella pneumoniae* in the neutropenic murine pneumonia model. J Antimicrob Chemother. 2022;77:1931-7.

[41] Mushtaq S, Garelo P, Vickers A, Woodford N, Livermore DM. Activity of ertapenem/zidebactam (WCK 6777) against problem Enterobacterales. J Antimicrob Chemother. 2022;77:2772-8.

[42] Drawz SM, Bethel CR, Hujer KM, Hurlless KN, Distler AM, Caselli E, et al. The role of a second-shell residue in modifying substrate and inhibitor interactions in the SHV beta-lactamase: a study of ambler position Asn276. Biochemistry. 2009;48:4557-66.

[43] Palzkill T. Structural and Mechanistic Basis for Extended-Spectrum Drug-Resistance Mutations in Altering the Specificity of TEM, CTX-M, and KPC beta-lactamases. Front Mol Biosci. 2018;5:16.

- [44] Tooke CL, Hinchliffe P, Bragginton EC, Colenso CK, Hirvonen VHA, Takebayashi Y, et al. beta-Lactamases and beta-Lactamase Inhibitors in the 21st Century. *J Mol Biol.* 2019;431:3472-500.
- [45] Papp-Wallace KM, Barnes MD, Taracila MA, Bethel CR, Rutter JD, Zeiser ET, et al. The Effectiveness of Imipenem-Relebactam against Ceftazidime-Avibactam Resistant Variants of the KPC-2 beta-Lactamase. *Antibiotics (Basel).* 2023;12.
- [46] Niu S, Chavda KD, Wei J, Zou C, Marshall SH, Dhawan P, et al. A Ceftazidime-Avibactam-Resistant and Carbapenem-Susceptible *Klebsiella pneumoniae* Strain Harboring bla(KPC-14) Isolated in New York City. *mSphere.* 2020;5.
- [47] Findlay J, Poirel L, Juhas M, Nordmann P. KPC-Mediated Resistance to Ceftazidime-Avibactam and Collateral Effects in *Klebsiella pneumoniae*. *Antimicrob Agents Chemother.* 2021;65:e0089021.
- [48] Sun L, Li H, Wang Q, Liu Y, Cao B. Increased gene expression and copy number of mutated bla(KPC) lead to high-level ceftazidime/avibactam resistance in *Klebsiella pneumoniae*. *BMC Microbiol.* 2021;21:230.
- [49] Maharjan RP, Sullivan GJ, Adams FG, Shah BS, Hawkey J, Delgado N, et al. DksA is a conserved master regulator of stress response in *Acinetobacter baumannii*. *Nucleic Acids Res.* 2023;51:6101-19.
- [50] Lan P, Lu Y, Liao W, Yu Y, Fu Y, Zhou J. Cefiderocol-resistant hypervirulent *Klebsiella pneumoniae* with CirA deficiency and co-production of KPC-2 and SHV-12. *Clin Microbiol Infect.* 2024.
- [51] Jousset AB, Poignon C, Yilmaz S, Bleibtreu A, Emeraud C, Girlich D, et al. Rapid selection of a cefiderocol-resistant *Escherichia coli* producing NDM-5 associated with a single amino acid substitution in the CirA siderophore receptor. *J Antimicrob Chemother.* 2023;78:1125-7.
- [52] McElheny CL, Fowler EL, Iovleva A, Shields RK, Doi Y. In Vitro Evolution of Cefiderocol Resistance in an NDM-Producing *Klebsiella pneumoniae* Due to Functional Loss of CirA. *Microbiol Spectr.* 2021;9:e0177921.
- [53] Nurjadi D, Kocer K, Chanthalangsy Q, Klein S, Heeg K, Boutin S. New Delhi Metallo-Beta-Lactamase Facilitates the Emergence of Cefiderocol Resistance in *Enterobacter cloacae*. *Antimicrob Agents Chemother.* 2022;66:e0201121.
- [54] Mezcord V, Traglia GM, Pasteran F, Escalante J, Lopez C, Wong O, et al. Characterisation of cefiderocol-resistant spontaneous mutant variants of *Klebsiella pneumoniae*-producing NDM-5 with a single mutation in cirA. *Int J Antimicrob Agents.* 2024;63:107131.
- [55] Chen H, Li N, Wang F, Wang L, Liang W. Carbapenem antibiotic stress increases bla(KPC) (-2) gene relative copy number and bacterial resistance levels of *Klebsiella pneumoniae*. *J Clin Lab Anal.* 2022;36:e24519.

■ CZA susceptible

▲ FDC susceptible

● FPZ susceptible

■ CZA resistant

▲ FDC resistant

● FPZ resistant

